

Edoardo Caciolo — Project Portfolio

Overview documentation for my projects. **All source code is private and proprietary.** These pages are deliberately high-level: they describe *what* each system does and the engineering behind it, but **not** the internal implementation, algorithms, or how to reproduce it. Code walkthroughs available to recruiters on request.

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Featured

- **ARBOR** — Multi-tenant AI platform that discovers, enriches, and reasons over real-world entities using a knowledge graph, vector search, and an event-sourced store.
- **Sentinel** — Multi-tenant document intelligence platform that extracts, scores, and attests risk in legal and financial documents via a verified multi-agent pipeline.
- **Tay** — Apple-platform app that recognizes physical art and monuments on-device and overlays real-time AR experiences anchored to them.
- **Kairo** — Gym-management platform with a Kotlin Ktor backend and native iOS/Android apps sharing a Kotlin Multiplatform core, covering access, bookings, subscriptions, lockers, and AI coaching.
- **people-os** — AI-assisted hiring platform exposing resume screening, interview scoring, and onboarding as MCP tools, with an eval framework measuring decision quality before changes ship.
- **fast-inference** — A self-hosted, OpenAI-compatible inference server for embedding, reranking, and generation models, built to keep RAG workloads fast and on-premise.
- **Anacleto** — A FastAPI platform that analyzes Instagram images, optimizes feed order with MCTS/Beam Search, schedules posts, and publishes via the Meta Graph API.

All projects

Project	What it is	Stack	Status
Sentinel	Multi-tenant document intelligence platform that extracts, scores, and attests risk in legal and financial documents via a verified multi-agent pipeline.	Python 3.12+, FastAPI, Uvicorn, Pydantic v2	beta
RLM Toolkit	A Python implementation of Recursive Language Models: an LLM solves tasks by writing and running REPL code and querying sub-models recursively, rather than answering in one pass.	Python 3.11+, FastAPI, Pydantic, rich	Beta
pdf-finder	Python library that locates text in PDFs and returns word-level bounding boxes, tolerant of OCR errors, typos, and page-spanning matches.	Python 3.11+, PyMuPDF (fitz), rapidfuzz, Smith-Waterman alignment	beta

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OCR Toolkit	Python library that extracts structured text, tables, formulas, and layout from PDFs using the dots.mocr vision-language model with a pluggable engine interface.	Python, dots.mocr (dots.ocr fork), PyTorch, HuggingFace transformers	beta
grounding-toolkit	Provider-agnostic Python engine that verifies whether LLM claims are grounded in their source documents and flags hallucinations, via a multi-tier verification cascade.	Python 3.11+, pydantic, httpx, PyYAML	alpha
exchequer-toolkit	Multi-jurisdiction fiscal engine that computes VAT, reverse charge, stamp duty, and withholding for SaaS billing using Decimal money and YAML-defined country rules.	Python 3.11+, PyYAML, Jinja2 (scaffold), Copier templates	beta
DSPy Toolkit	A domain-agnostic library for compiling, validating, and optimizing DSPy modules from JSONL training datasets.	Python 3.11+, DSPy, LiteLLM, Typer	beta
cognis-toolkit	A consumer-agnostic Python knowledge engine composing a pluggable LLM router, RAG retrieval, a wiki domain model, and hybrid PII detection behind injected Protocol interfaces.	Python 3.11+, Pydantic 2, PyYAML, HTTPX	beta
Aptus	Career-intelligence platform that aligns a CV with a job description and returns a scored, evidence-backed, auditable match.	Python 3.12, FastAPI, LangGraph, LiteLLM	prototype
Mandrake	A multi-agent AI trading platform that keeps a human operator in the loop and enforces hard, non-bypassable risk constraints on every trade.	Python, FastAPI, Uvicorn, Typer	Pre-alpha (Phase 0 foundation)

Project	What it is	Stack	Status
ARBOR	Multi-tenant AI platform that discovers, enriches, and reasons over real-world entities using a knowledge graph, vector search, and an event-sourced store.	Python 3.12, FastAPI, Pydantic v2, PostgreSQL 16 + PostGIS	Active private project / production-oriented; backend v1.1.0, proprietary license
Sentinel Red	AI security platform that classifies documents to NATO/Gov standards with a no-downgrade deterministic layer and red-teams LLM endpoints mapped to MITRE ATLAS.	Python 3.11+, FastAPI, Uvicorn, Pydantic v2	production (v1.0.0)
ClusterMind	Deterministic, auditable CLI for capacity planning and cloud-cost estimation of web/SaaS and self-hosted LLM services.	Python 3.11-3.13, Typer, Rich, PyYAML	beta
BOT Garage	Local-first FastAPI app for motorcycle/vehicle maintenance tracking with on-device OCR import of registration and invoice PDFs.	Python, FastAPI, Uvicorn, Pydantic	Working personal application, actively developed (single-owner oriented)
Tay	Apple-platform app that recognizes physical art and monuments on-device and overlays real-time AR experiences anchored to them.	Swift 6, SwiftUI, ARKit, RealityKit	Early development / prototype
Kairo	Gym-management platform with a Kotlin Ktor backend and native iOS/Android apps sharing a Kotlin Multiplatform core, covering access, bookings, subscriptions, lockers, and AI coaching.	Kotlin, Swift, Kotlin Multiplatform, Ktor	beta
people-os	AI-assisted hiring platform exposing resume screening, interview scoring, and onboarding as MCP tools, with an eval framework measuring decision quality before changes ship.	Python, Anthropic Claude API, FastMCP (Model Context Protocol), LangGraph	Prototype (v0.1.0)

Project	What it is	Stack	Status
fast-inference	A self-hosted, OpenAI-compatible inference server for embedding, reranking, and generation models, built to keep RAG workloads fast and on-premise.	Python, Triton, ONNX Runtime GPU, ONNX	Prototype / portfolio engineering project (v0.1.0), MIT License
Anacleto	A FastAPI platform that analyzes Instagram images, optimizes feed order with MCTS/Beam Search, schedules posts, and publishes via the Meta Graph API.	Python 3.11, FastAPI, Uvicorn, Pydantic	beta
NEMESIS	Design specification and IEEE research paper for a three-agent adversarial AI security system where agents attack, defend, and evolve against a codebase.	Markdown, LaTeX (IEEE, EN/IT), HTML/CSS (GitHub Pages), Claude (Anthropic API) [designed]	Pework / design specification (no implementation committed)
Hybrid Multi-Layer Control for Robust Autonomous Driving	Bachelor's thesis plus a Python digital-twin framework implementing a layered RL-MPC control stack for a Formula Student driverless car under uncertain tire grip.	Python, MATLAB, LaTeX, Stable-Baselines3 (PPO)	Active academic project (thesis, v0.0.1) — research prototype
NETWATCH	Linux endpoint security monitor that uses eBPF to watch kernel-level process, syscall, network, and file activity, then detects attacks with rules and ML and dispatches alerts to a dashboard and SIEM.	Python, C (eBPF), JavaScript (React/JSX), eBPF / BCC	Beta
Porfirio Magazine	Full-stack digital magazine platform with bilingual publishing, Stripe payments, engagement analytics, and AI-driven Instagram post generation.	React 18, Vite, JavaScript (JSX), Tailwind CSS	Production (deployed on Render, actively maintained; no LICENSE file present)
MINERVA	Multi-source OSINT engine that collects public threat data, correlates it in a Neo4j knowledge graph, and answers natural-language threat-intelligence questions.	Python 3.11+, FastAPI, Uvicorn, Pydantic v2	Beta

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Basil	A Python CLI that generates printable crossword puzzles from a word list and hides a secret phrase in the grid for scavenger hunts.	Python, Pillow (PIL), Standard library (json, os, random, copy), Interactive terminal CLI	prototype
Tappo	An iOS counter and goal tracker with multi-project history, streaks, heatmaps, Home Screen widgets, and Lock Screen Live Activities, built on SwiftUI and SwiftData.	Swift, SwiftUI, SwiftData, WidgetKit	prototype
LitInvestor Blog	Full-stack financial-literacy blog with article management, reader accounts, and AI-assisted generation of Instagram carousel images from articles.	Python 3.11, Flask, SQLAlchemy, Alembic	Production
marr-ml-projects	Two graduate ML coursework projects: a six-classifier tabular benchmark and a CNN controller for the Gymnasium CarRacing-v2 simulator.	Python, Jupyter Notebook, scikit-learn, XGBoost	Academic coursework (completed assignments); no license
Interactive Graphics Projects	Six self-contained, browser-based computer graphics assignments implementing core rendering and simulation techniques from scratch in JavaScript and GLSL.	JavaScript, GLSL, HTML, WebGL	Completed coursework